

- Q.34 Write a short note on detector.
Q.35 Explain water Pollution Monitoring system.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Explain fundamental blocks of Analytical Instruments.
Q.37 Draw & Explain thermal conductivity Analyzer with their Applications.
Q.38 Explain NMR Spectroscopy in detail with each element are used in this spectroscopy.

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5th Sem / Instrumentation & Control, EI Sub : Analytical & Environmental Instruments

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Cell is a primary cell.
a) True b) False
- Q.2 The thermistor whose resistance increase by increasing temperature.
a) NTC thermistor b) PTC thermistor
d) PTC thermistor d) Any of these
- Q.3 PH of neutral salt.
a) 7 b) <7
c) >7 d) 0
- Q.4 Electrolytes conduct electric current.
a) By the movement of ions
b) By the movement of atoms
c) By the movement of molecules
d) By the movement of electrons from the cathode to anode
- Q.5 Which of the following is the formula of PH calculation?
a) $\text{Log } 10 [\text{H}^+]$ b) $-\text{Log } 10 [\text{H}^+]$
c) $\text{Log } 2 [\text{H}^+]$ d) $-\text{Log } 2 [\text{H}^+]$

- Q.6 Mostly pollution of rivers takes place by discharge sewage.
 a) True b) False
- Q.7 Pure water is known to be which of the following:
 a) Weak electrolyte
 b) Strong electrolyte
 c) Neither weak nor strong
 d) Non an electrolyte
- Q.8 Which of the following is not used for detection in gas chromatography.
 a) Infrared spectroscopy
 b) NMR
 c) Flame ionization
 d) Electrical Conductivity
- Q.9 LED is a display device.
 a) True b) False
- Q.10 In mass spectrometer, the sample that has to be analyzed in bombarded with which of the following.
 a) Protons b) Electrons
 c) Neutrons d) Alpha particles

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Explain IOT.
 Q.12 What is chromatography?
 Q.13 Define Pollution.
 Q.14 Column.

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- Q.15 What is display unit?
 Q.16 What is Ph?
 Q.17 Noise
 Q.18 Spectroscopy.
 Q.19 Define Cell.
 Q.20 LED.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 What is Chromatography and give their name of types of chromatography.
 Q.22 What do you mean by Liquid Analysis?
 Q.23 Describe the Various blocks of Analytical Instruments.
 Q.24 How air quality system can be monitored using IOT.
 Q.25 Explain Injector Instruments of GAS Chromatography.
 Q.26 Explain Mass Spectroscopy of brief.
 Q.27 Write a short note on Electro-Chemical Analyzer.
 Q.28 Explain working Principle of Infrared Gas Analyzer.
 Q.29 How Arduino can be used for monitoring the quality of water pollution.
 Q.30 Discuss the role of column used in chromatography.
 Q.31 Write a short note on noise Pollution.
 Q.32 Explain conductivity meter.
 Q.33 Write a short note on electrodes used for Ph measurement.

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